

5-2-1: Examining Network Routers

At the end of this episode, I will be able to:

1. Identify and explain the fundamental role and functionality of network routers, given a scenario.

Learner Objective: *Identify and explain the fundamental role and functionality of network routers, given a scenario*

Description: In this episode, the learner will examine the functionality and role network routers perform in modern networking communications. We will explore router functions, operational principles and benefits.

- What is a router and its role in networks?
 - o Efficient and timely routing of packets, through path selection.
 - o Connect multiple networks together.
 - o Route data packets between networks, using IP addresses.
 - o Operate at the network layer (Layer 3) of the OSI model.
 - o Use IP addresses to make decisions about the best path for data transfer.
 - o Isolation of broadcast domains.
- What are routers doing in networks?:
 - o Path selection:
 - ♣ Routers use IP addresses and routing protocols to determine the most efficient path for data packet traversal across networks.
 - o Routing tables & route selection:
 - ♣ A routing table contains a list of routes to various network destinations
 - ♣ This information is used to decide the best path for forwarding packets, based on the destination IP address.
 - o Network Address Translation (NAT):
 - ♣ A process that translates private IP addresses to a public address for internet communication.
 - ♣ Conserves public IP addresses.
- What are some benefits of routers?
 - o Connecting networks:
 - ♣ Enable communication between different topologies and architectures.
 - ♣ Examples: LAN to WAN, LAN to Internet.
 - o Internet access:
 - ♣ Connect local networks to the internet and manage data traffic efficiently.
 - o Broadcast communications isolation:
 - ♣ Segment broadcast domains.
 - ♣ Isolate broadcast communications to specific segments.
- What are some comparisons to other network appliances?
 - o Switches:
 - ♣ Data link layer (Layer 2).
 - ♣ Connect devices within the same network.
 - ♣ Use MAC addresses.
 - o Hubs:
 - ♣ Physical layer (Layer 1).
 - ♣ Basic devices.
 - ♣ Broadcast to all ports.
 - o Routers:
 - ♣ Network layer (Layer 3).
 - ♣

- Are more intelligent.
 - ♠ Connect different networks.
 - ♠ Make decisions based on logical IP addresses.
 - ♠ Do not pass broadcast traffic across networks.
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- Additional Resources:
 - o If applicable